Basics of

Brain-Computer Interface

AI-Driven Sri Lanka - technical discussion

Isuru Jayarathne
Postdoctoral researcher
University of Aizu, Japan
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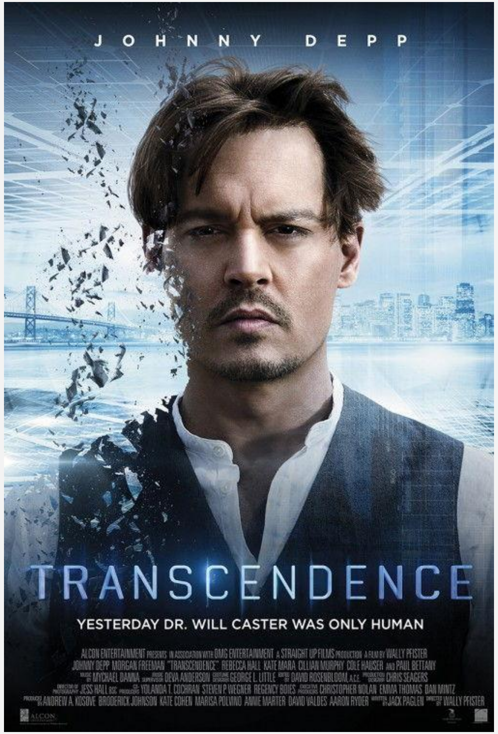
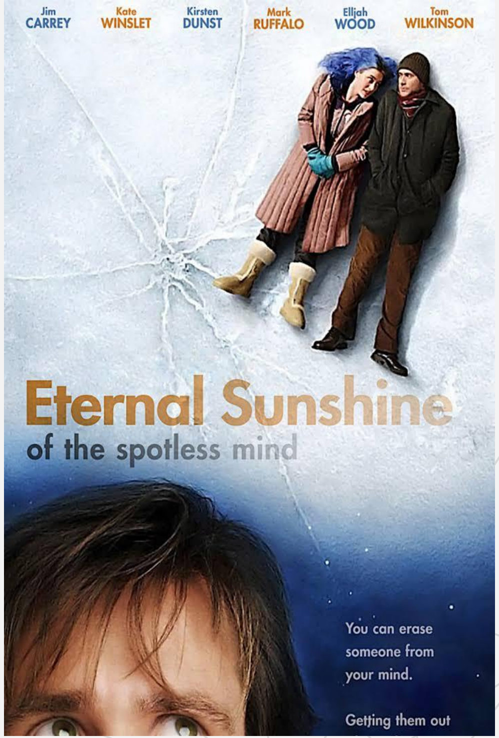
Contents

Think beyond the reality





BCI applications

- Controlling robots, 2017

BCI applications

- Controlling drones, 2015

BCI applications

- BCI speller and mouse, 2011

BCI applications

- Image reconstruction using brain data, 2017

BCI applications

- Brain-to-vehicle by Nissan, 2018

BCI applications

- BCI with VR, 2016

BCI applications

- BCI exoskeleton, 2015

BCI applications

- BCI in field of education

BCI applications

- Brain signal-based authentication system
- Epilepsy seizure detection
- Meditation applications
- Emotion classification
- Wheelchair control
- Sleep disorder studies
- etc.

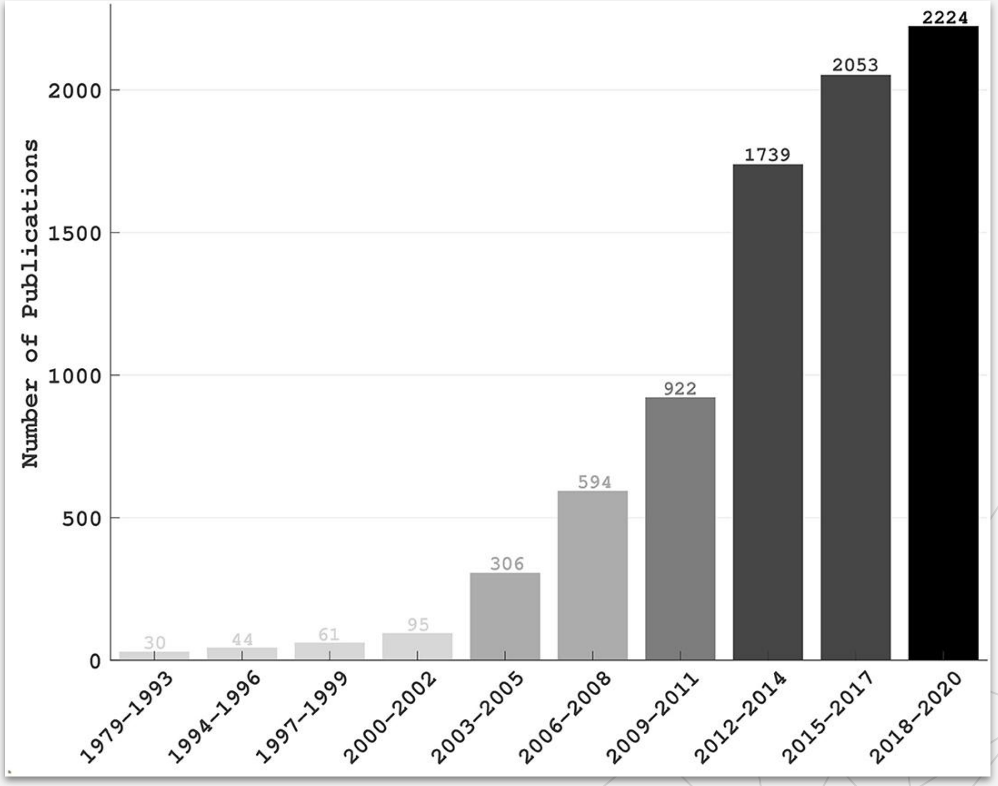
Popularity

NEURALINK

facebook

NISSAN

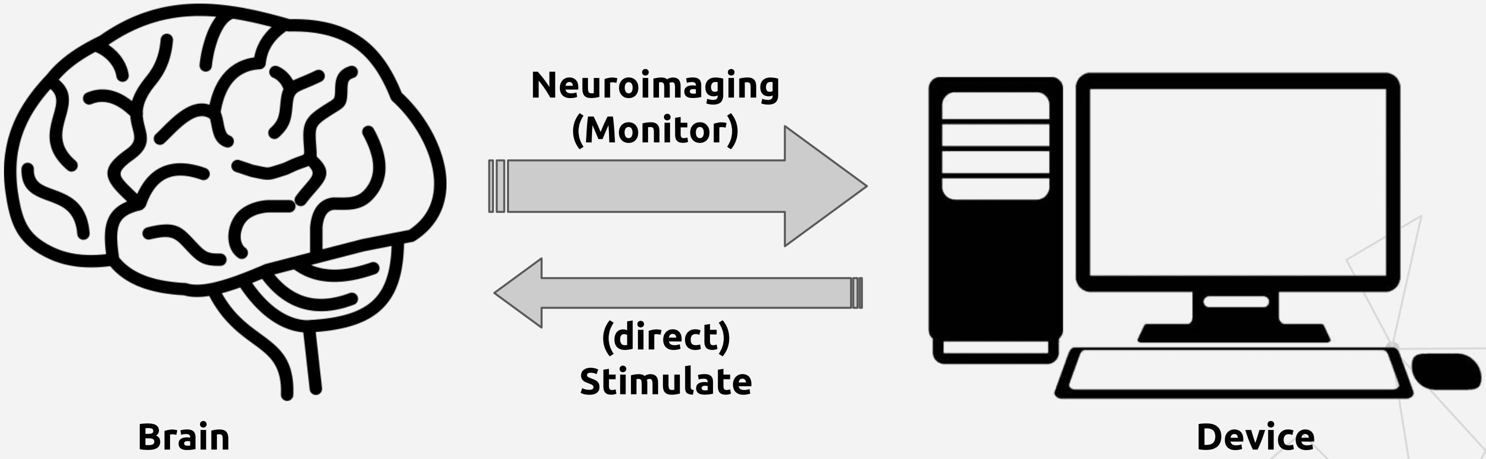
Microsoft



Source: https://doi.org/10.3389/fnsys.2021.578875

What is BCI?

- Direct communication between enhanced brain and an external device.
- Subcategory of Human-computer interaction (HCI)



Related subject areas

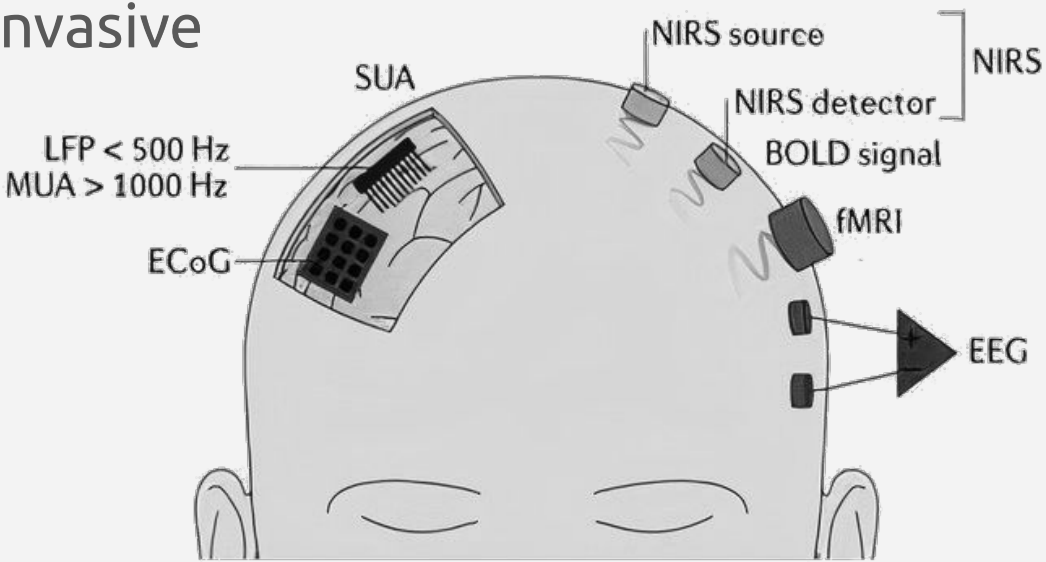
- **Brain anatomy**: study of brain structure
- **Neuroscience**: study of structure or function of the nervous system and brain

What computer scientist/engineers familiar with

- **Digital signal processing (DSP):** process signals digitally using computing devices
- **Digital image processing (DIP):** process images digitally using computing devices
- **Machine learning (ML)**: ability to find patterns in data

Interacting with brain

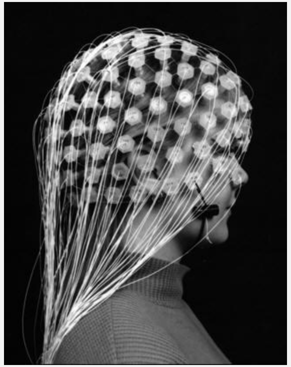
- Methods
  - Non-invasive
  - Partially invasive
  - Invasive

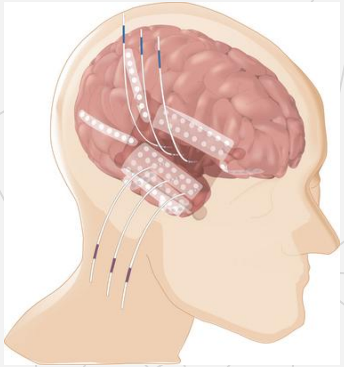


Source: https://www.nature.com/articles/nrneurol.2016.113

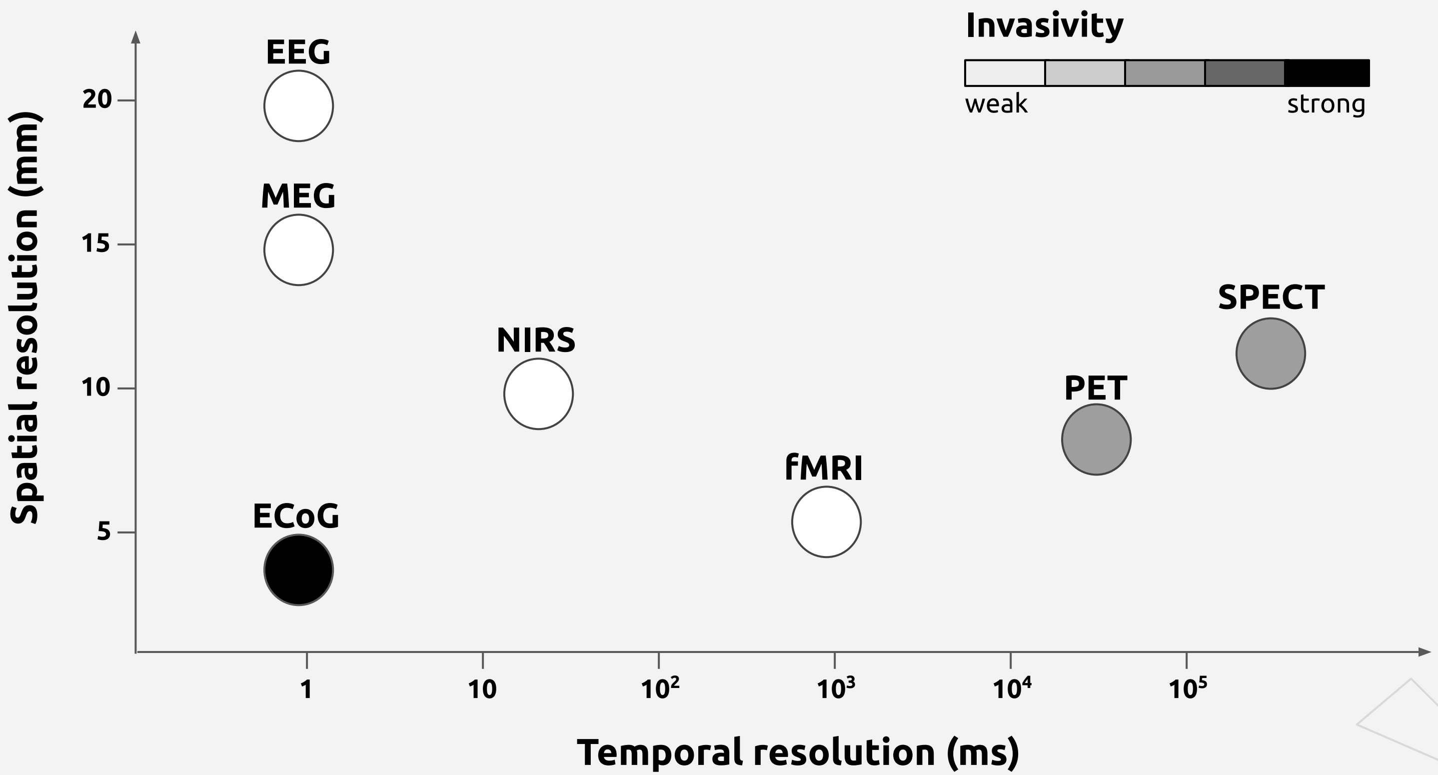
Brain activity monitoring techniques

- **fMRI**: Functional Magnetic Resonance Imaging
- **EEG**: Electroencephalography
- **MEG**: Magnetoencephalography
- **NIRS**: Near-infrared spectroscopy
- **ECoG**: Electrocorticography

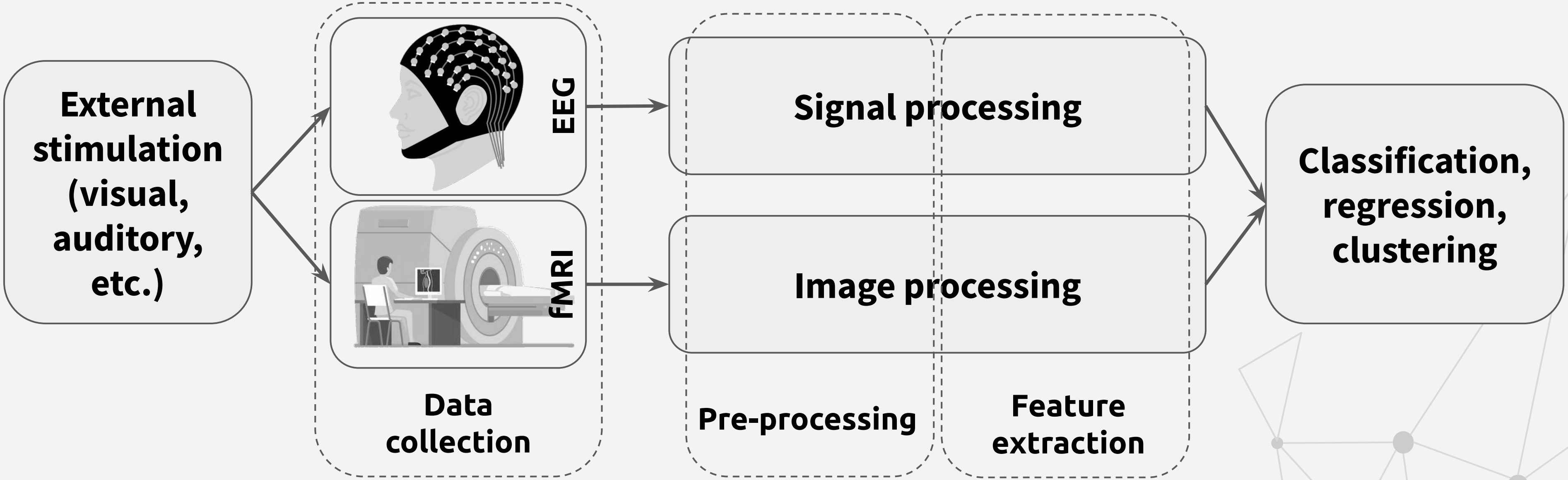



Spatial and temporal resolution of neuroimaging



Source: https://tel.archives-ouvertes.fr/tel-01175851/

Basic flow of most BCI systems

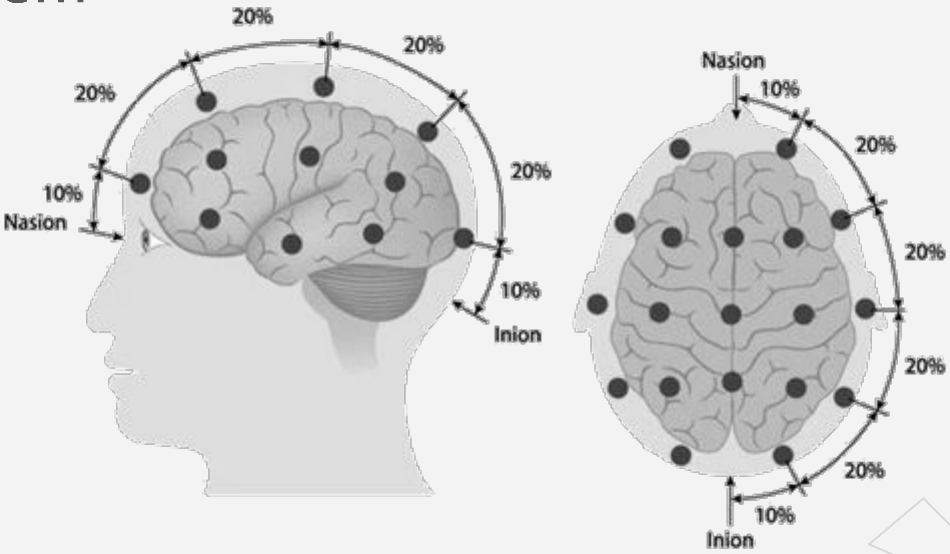


What is EEG and why EEG?

- Electroencephalography (EEG): Electrophysiological monitoring method to record electrical activity of one's brain
- Subtle electric field is generated by neurons
- EEG electrodes can capture the electric field of group of neurons
- Portable, non-invasive, affordable, and easy to use

Electrode placements of EEG systems

- 10-20 system
- 10-10 system or 10%
- 10-5 system



Source: https://doi.org/10.1186/s13634-015-0251-9

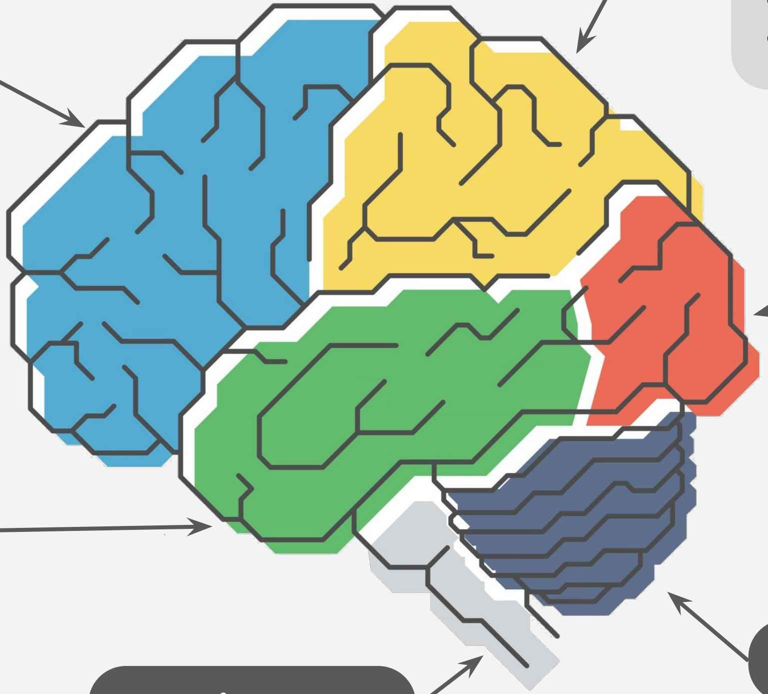
Brain lobes

Frontal lobe

- Problem solving
- Emotional traits
- Reasoning
- Personality

Parietal lobe

- Sensation
- Language
- Intelligence
- Reading



Occipital lobe

- Vision
- Colour
- Perception

Temporal lobe

- Hearing
- Memory
- Understanding
- Behavior

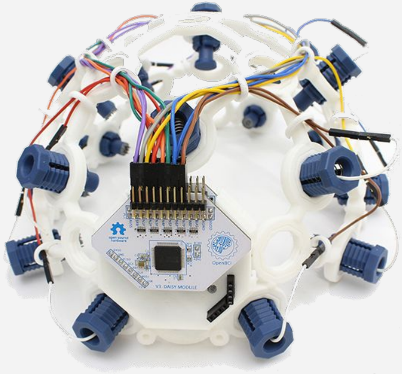
Brain stem

Cerebellum

EEG devices

- Clinical- and consumer-grade devices are available
- Two types of electrodes, Wet and dry
- Has different number of electrodes

OpenBCI

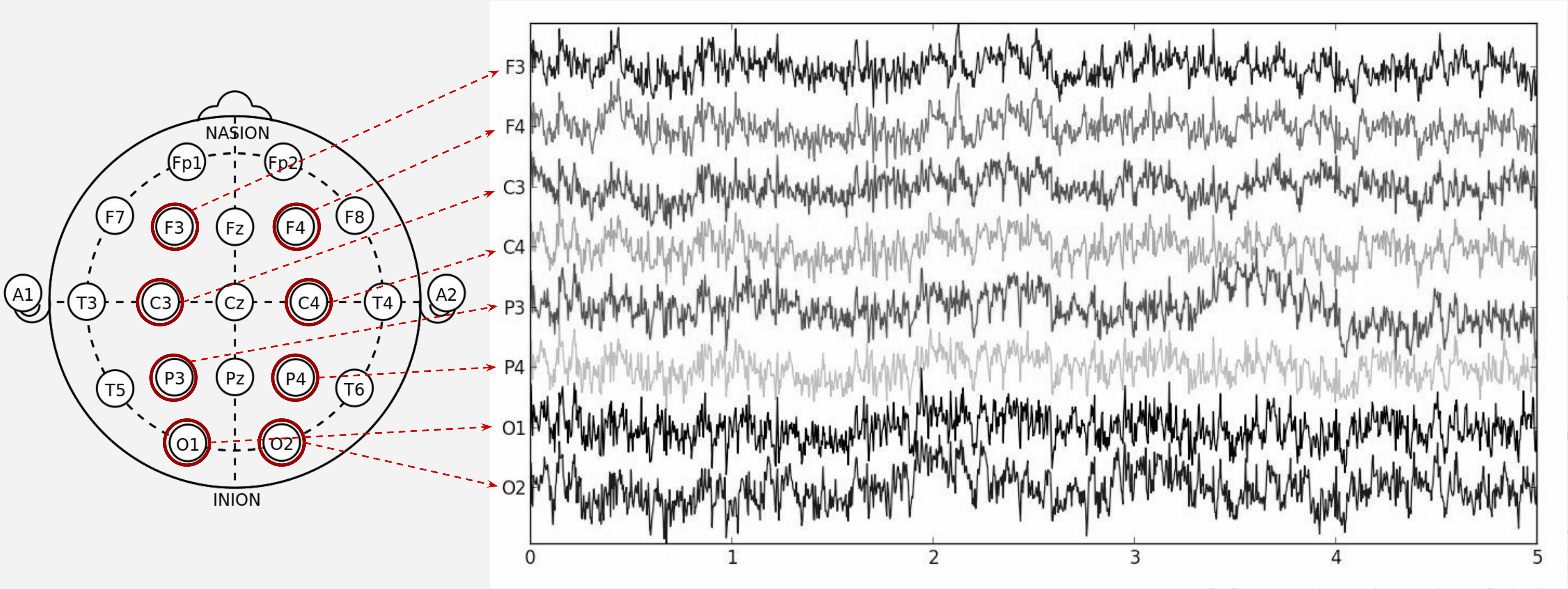


Emotiv EPOC+

Emotiv Insight

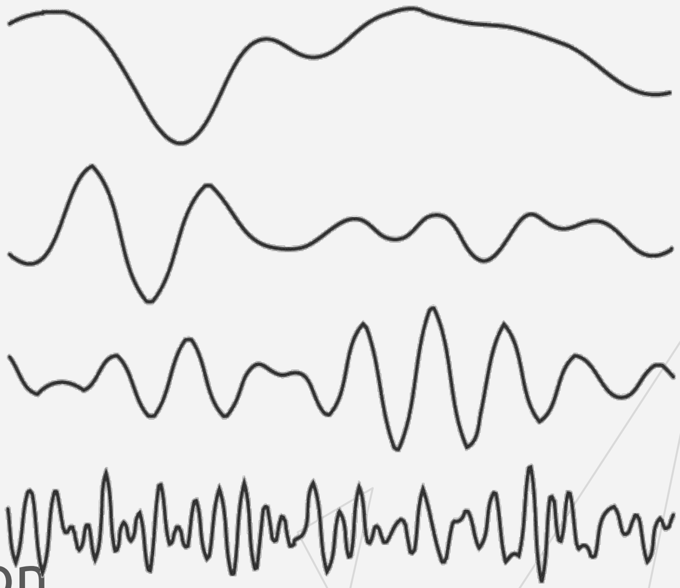
Muse

Sample EEG signal



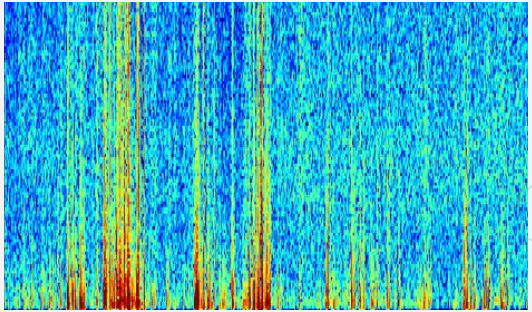
EEG frequency bands

- Delta: 1 - 4 Hz
  - Highest amplitude
  - Presents during deep sleep
- Theta: 4 - 8 Hz
  - Mental and physical relaxation
- Alpha: 8 - 13 Hz
  - Focused attention, memory recall
- Beta: 13 - 30 Hz
  - Active thinking, active concentration



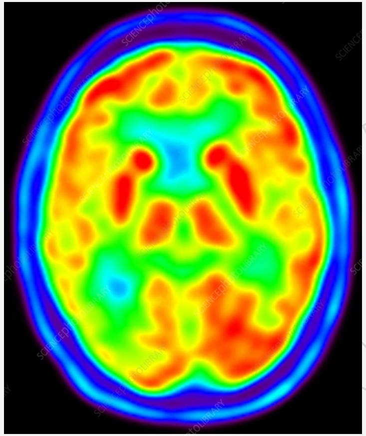
Pre-Processing

- Artefact (noise) removal
  - ECG, eye blinking, power-line noise
  - Blind Source Separation (BSS)
- Filtering (low-pass, high-pass, band-pass, notch)
  - Separate frequency bands
- Domain transform
  - Fourier transform (time to frequency domain)
  - Spectrogram
  - Periodogram



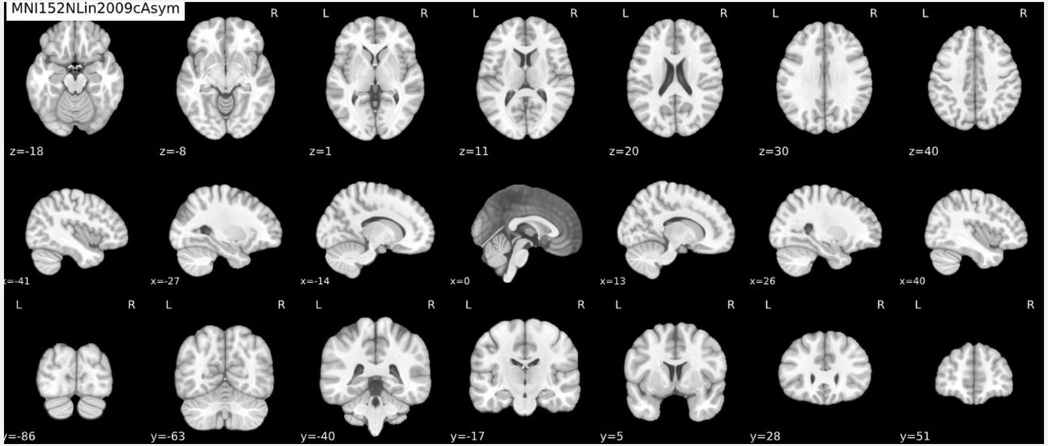
fMRI and PET

- fMRI: measures brain activity by detecting changes associated with blood flow
- Positron Emission Tomography (PET): visualize and measure changes in metabolic processes (blood flow, regional chemical composition, and absorption)



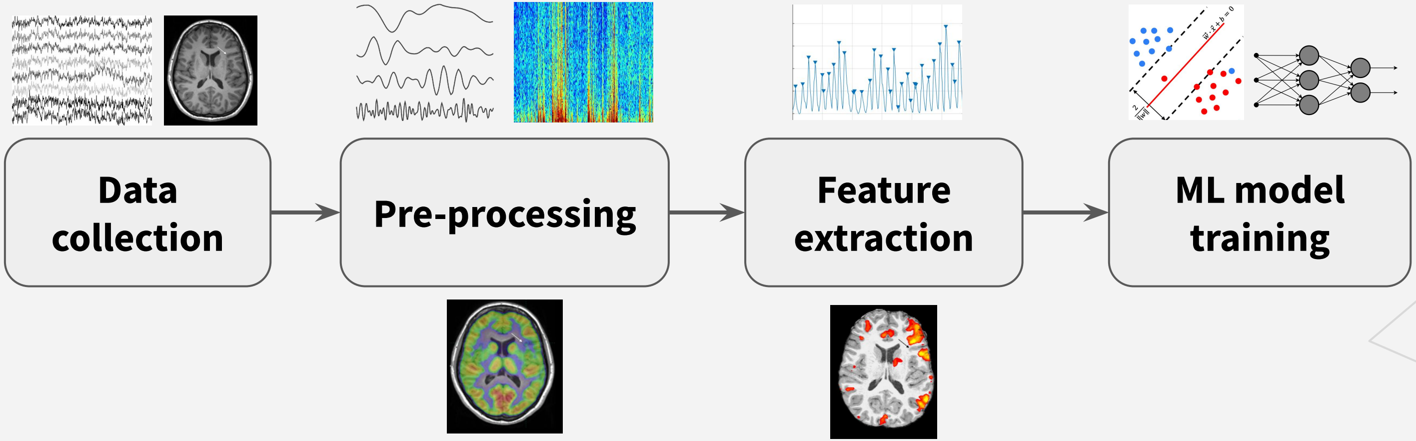
fMRI and PET (cont.)

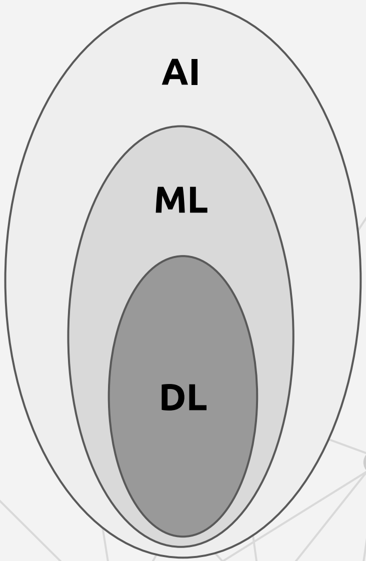
- High spatial resolution and low temporal resolution
- Highly used in brain pathology (study of the causes and effects of disease or injury)
- Output is a 2D image of given plane of the 3D brain
- Scanners are not portable, expensive



Role of conventional machine learning in BCI

- Traditional ML techniques (LDA, SVM, kNN, ANN)
  - Feature extraction is needed
  - Low computational cost
  - Even small dataset can be used





Feature extraction methods for EEG

Time-domain features

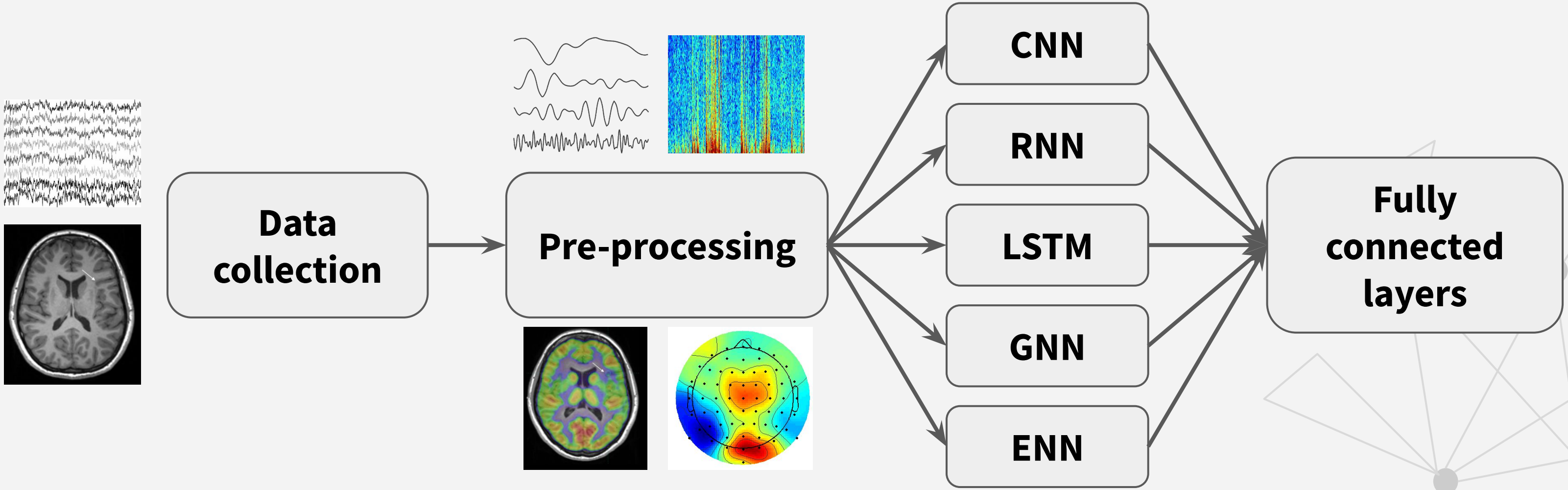
- Common spatial patterns (CSP)
- Autoregressive coefficients (AR)
- Slope sign change (SSC)
- Zero crossing
- Waveform length (WL)
- Willison amplitude
- Inter-hemispheric amplitude ratio (IHAR)

Frequency-domain features

- Power spectral density (PSD)
- Peak frequency.
- Power spectrum deformation
- Frontal asymmetry
- Spectral coherence

Role of deep learning in BCI

- No need to extract features
- Need a large dataset
- High computational cost
- Transfer learning can be used in some cases

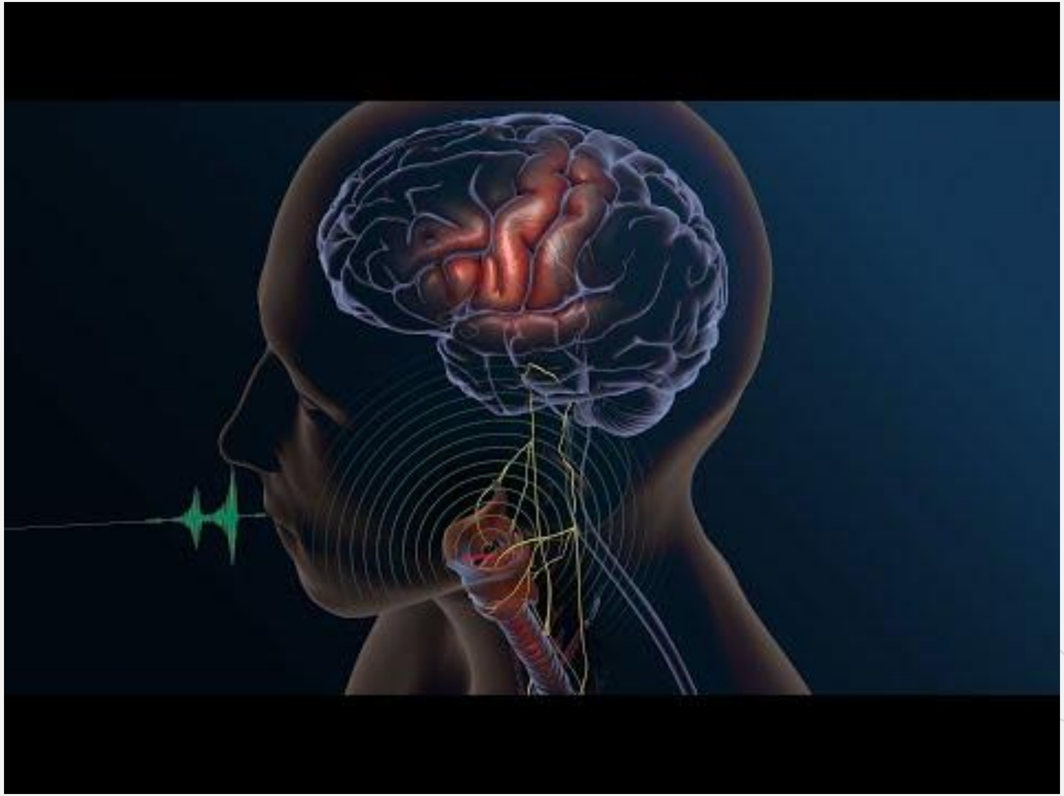


Challenges

- Uncertainty of brain patterns
- Difficulty of data collection
- Unreliability of data quality
- Misbelief about BCI technologies

Most recent BCI applications

- Speech from brain signals, 2019



Most recent BCI applications

- Brain-to-text system, 2021

Can we control brain with a device?

- Repetitive Transcranial Magnetic Resonance (rTMS)
- Transcranial Direct Current Stimulation (tDCS)

Most recent BCI applications

- Effect of rTMS on speech, 2011

Big picture

- BCI
  - Neuroimaging
    - Functional
      - fMRI
      - EEG
      - fNIRS
      - ECoG
    - Molecular
      - PET
      - SPECT
      - mMRI
    - Structural
      - CT
      - MRI
  - Stimulate
    - Direct
      - Electrical
        - tDCS
      - Magnetic
        - rTMS
      - Chemical
        - Psychoactive drugs
    - Indirect
      - Visual
      - Auditory
      - Taction
      - Gustation
      - Olfaction

Recreating brain

- Baby X

Thank you!

Any questions?

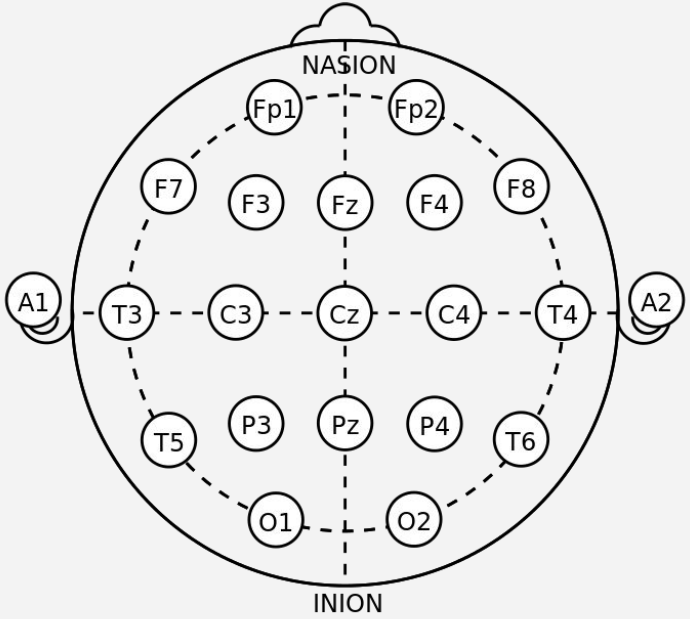
Blog: www.i-linkz.net

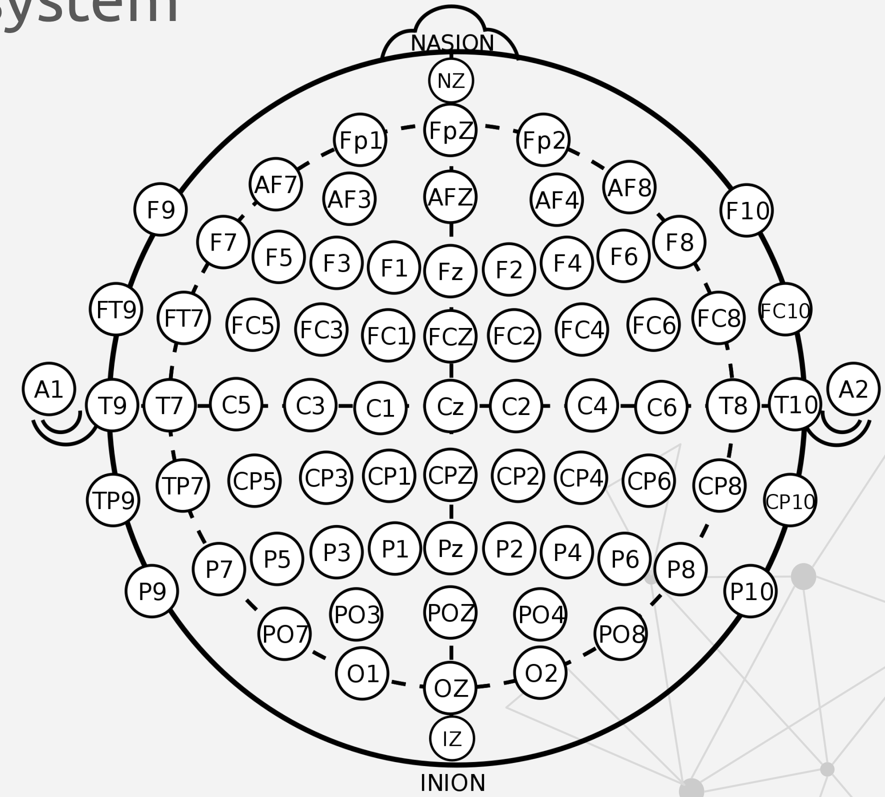
Biosignals

- Signals that can be continually measured and monitored from the living being
- Can be electrical or non-electrical
  - Electroencephalogram (EEG)
  - Magnetoencephalogram (MEG)
  - Electrocorticography (ECoG)
  - **Electrocardiogram (ECG)**
  - Electromyogram (EMG)
  - Electrooculography (EOG)
  - Galvanic skin response (GSR)
  - Mechanomyogram (MMG)

Electrode placement (Cont.)

- 10-20 system & 10% system





EEG devices (Cont.)

Device	Number of Channels	Electrode type
NeuroSky Mindwave	1	Dry
Muse	2	Dry
EMOTIV Insight	5	Dry
EMOTIV Epoc+	14	Wet
OpenBCI	16	Dry
mBrainTrain	24	Wet
ENOBIO 32	32	Dry
Cognionics Mobile- 72	64	Dry

Artifacts

- EOG, EMG, ECG interferes the EEG signals
- Impact of power line noise
- Head swaying and swinging

To remove

- Record other biosignals as well
- Filtering
- Record gyroscope data

Other interesting stuffs

- rTMS, tDCS
- Neuromorphic chips
- Quantum Computers (ibm.biz/qx-introduction)
- Augmented Reality (AR) and Virtual Reality (VR) with HMDs and smart glasses

Contents

References

- IMOTION EEG Pocket Guide (url: https://imotions.com/eeg-guide-ebook/)
- https://www.youtube.com/watch?v=Zd9WhJPa2Ok
- https://youtu.be/hLjxMjBlB9k
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- www.emotiv.com
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